



Visualizing distances as a function of speed: Design and evaluation of a distance-speedometer

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ABSTRACT

This paper contributes to the research on transportation and human factors by designing and evaluating an ecological human-machine interface for speed and distance control. In the future, traffic management will determine optima for speed and safety distances. Wirelessly broadcasted, the question remains how to present this information to drivers in a way that is easily understood and intuitively followed. To this end, a human-machine interface, called distance-speedometer, was first developed according to the principles of ecological interface design and then evaluated in a driving simulator study with forty-nine participants. It presents augmented distances derived from speed in a head-up display. Using a within-subject design, the distance-speedometer and standard speedometer were compared against each other in two scenarios (car-following & sign-following). Each scenario triggered several speed changes. When reacting to speed signs, driving performance did not differ between the two interfaces. However, with the distance-speedometer, visual workload was reduced as more secondary tasks could be completed. When reacting to speed changes of a car ahead, the engagement in the secondary task did not differ significantly between the two interfaces. However, the variance of speed deviations was significantly lower with the distance-speedometer. Results can be explained by the ability to process the distance-speedometer in a skill-based manner, whereas the normal speedometer demands rule-based behaviour to control the speed. With regard to traffic management, the distance-speedometer might be used to unconsciously influence the driver's choice of speed in order to increase traffic safety and efficiency.

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1. Introduction

Given their impact on safety and efficiency, a driver's choice and control of both speed and following distance are essential factors in road traffic. Maintaining an appropriate speed avoids a disproportionate increase in the risk of crashes (Kloeden, Ponte, & McLean, 2001) and their severity (Elvik, Christensen, & Amundsen, 2004; Kröyer, 2015). More homogeneous driving behavior leads to an increased traffic flow (Brilon, Geistefeldt, & Regler, 2005) and correlates with fewer accidents (Finch, Kompfner, Lockwood, & Maycock, 1994). Therefore, speed management is of major interest in traffic management (Marchau, van Nes, Walta, & Morsink, 2010) and serves the following two main goals:

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1. The determination of set values with regard to speed and distance.
2. The application of measures that cause drivers to adopt the recommendations.

Concerning the second goal, it is helpful to look at general descriptions of the driving task. As [Rasmussen \(1983\)](#) has pointed out, there are three levels of behaviour, which rely on skill-based, rule-based and knowledge-based cognitive control in an ascending order of consciousness. Basic driving, e.g. steering along the street, is mostly skill-based and thus automatic and largely effortless. In a changing environment, with e.g. street signs, learned rules (if-then relationships) apply. This rule-based behaviour is more conscious and therefore less smooth and less automated than skill-based behaviour. Lastly, knowledge-based behaviour applies when conscious problem solving is necessary. More conscious tasks, such as reading the speedometer, utilize focal visual resources, while skill-based processes like steering along the street also utilize ambient vision ([Horrey & Wickens, 2004](#)). In accordance with [Wickens \(2002\)](#) model of multiple resources, automated behaviour allows for parallel focal tracking of e.g. safety critical stimuli. Regarding cognition, automated behaviour has the advantage that resources for more conscious behaviour, like coping with hazards, stay free ([Wickens & Hollands, 2000](#)). The approach to design human-machine interfaces, which allow for behaviour based on the lowest cognitive control level possible (skill-based behaviour), is called ecological interface design approach ([Vicente & Rasmussen, 1992](#); [Vicente, 2002](#); [Burns & Hajdukiewicz, 2004](#)).

With regard to the maintenance of recommended speed, typical means of traffic management rely on rule-based behaviour. Primarily, the recommended speed is displayed either on static or dynamically changing traffic signs ([Smulders, 1990](#)). Since empirical research suggests that speeding and tailgating commonly occur unintentionally ([Schmidt & Tiffin, 1969](#); [Matthews, 1978](#); [Taieb-Maimon & Shinar, 2001](#); [Ben-Yaacov, Maltz & Shinar, 2002](#)), various in-vehicle systems were developed as countermeasures ([Várhelyi, 2002](#)). They may simply inform, such as by showing the current speed limit next to the speedometer, or they may issue a warning if the car is going too fast or if it is too close behind the car ahead ([Brookhuis & de Waard, 1999](#); [Maltz & Shinar, 2004](#)).

What is common to all these approaches is that the driver first has to understand that there is a difference between the recommended speed (the set value) and his current speed (the control variable), and he then has to decide to control his speed accordingly (rule-based). Thus, it would be preferable to present the recommended speed to be maintained in a manner that allows for skill-based control. Such information about the appropriate speed should directly trigger speed control behaviour, while being less demanding. This kind of skill-based control is also useful when it comes to intentional violations. Studies on driver assistance systems, like the intelligent speed adaptation system (where the electronics force a certain speed reduction), have found that the drivers who are most likely to violate speed limits are the least likely to use these systems ([Jamson, 2006](#)). However, if the recommended speed were to be presented in a manner that triggered a skill-based control, drivers might follow the recommendations, as no conscious rule-based decision would be required.

In order to allow for skill-based control, an ecological interface makes the constraints and relationships of behaviour perceptually evident ([Rasmussen & Vicente, 1989](#); [Seppelt & Lee, 2007](#)). Concerning lateral vehicle control, it is the street itself that sets the visually perceptible boundaries. In the case of speed control, such continuous boundaries are not available, making unintentional violations of safe driving more likely. Only cars ahead, driving optimally, might constitute such boundaries. The same applies for optimal following distances which are not directly perceivable in the environment. They have to be estimated or calculated from counting seconds. In this regard, [Norman \(1986\)](#) points out that humans have goals that are psychologically rather than physically relevant and [Groeger \(2000\)](#) argues that distance information provided by electronics should be transformed into a more relative form. With regard to computed braking distances, [Bubb \(1975\)](#) as well as [Tönnis, Lange, and Klinker \(2007\)](#), used a contact analogue head-up display (HUD) to visualize them. They argue, that an augmented braking distance functions as a control variable and therefore enhances speed control. Nevertheless, no reference to a preferable distance (set value) was given in either study.

On the basis of the above theory, a human-machine interface (HMI) that integrates speed and distance was conceptualized using a contact analogue HUD. [Fig. 1](#) shows this so-called distance-speedometer when sign-following (top row) and car-following (bottom row). The columns indicate what this interface looks like when driving too slowly, at a suitable speed, and too fast.

As [Fig. 1](#) indicates, the current speed is visualized as a semi-transparent carpet that ends in a thick horizontal bar (originally blue). The carpet's length and the position of the bar are computed to visualize a preferable time headway (THW). This means, that the bar is closely behind the lead car when the driver is keeping a safe distance. When the driver is following too closely, the bar pushes under the car ahead. If the distance is too large, there is a gap between the bar and the car ahead.

When there is no car ahead or when it is going faster than recommended, the HMI works in the same manner. A smaller longitudinal bracket (originally white) is introduced as reference. The recommended speed is being driven when the bar lies within the bracket. If the driver is going too fast, the bar ends up in front of the bracket. In the case of the driver going too slowly, the bar falls behind the bracket. If the speed-related distance lags behind the bracket, the driver may be induced to drive faster in order to catch up. Similarly, the driver may be induced to fill in the distance between his car and the car ahead. Therefore, in addition to prevent speeding and tailgating, the distance-speedometer may also discourage driving too slowly and following at a distance that is too great for optimal traffic flow. This effect could be especially interesting for traffic management systems, which try to optimize traffic flow by specifically influencing the speed of vehicles. For example, the distance-speedometer theoretically should be capable of slowing a car down before it reaches traffic lights, dangerous crossings or roundabouts. Overall, by using the distance-speedometer, drivers get integrated information about speed and dis-

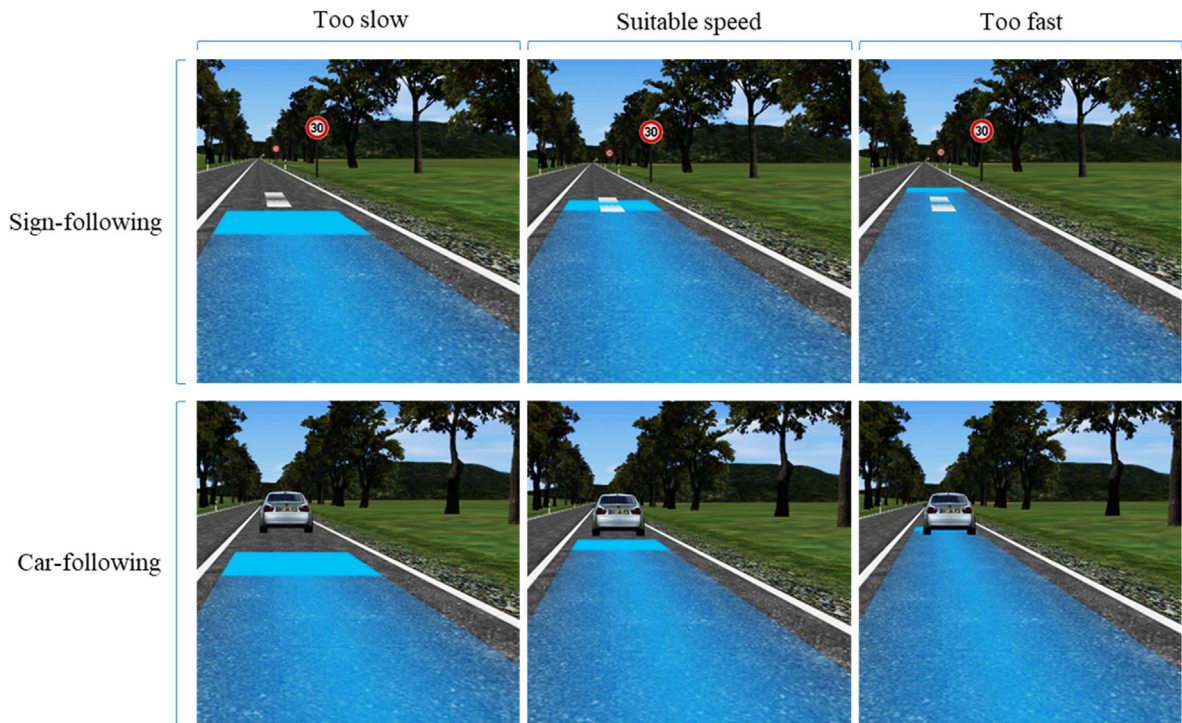


Fig. 1. Distance-speedometer in the simulated head-up display of the driving simulator. The current speed is visualized as a semi-transparent carpet that ends in a thick horizontal bar (originally blue) and this bar should be within the bracket (originally white). In the car-following scenario (bottom row) the car sets the limit and the thick horizontal bar should stay shortly behind. The figures' contrast was enhanced for better readability in grey-scale printouts. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

tance, which can be directly transferred into an adequate reaction such as slowing down, because theoretically skill-based behaviour is triggered.

In this paper, we evaluated the distance-speedometer in comparison to a standard speedometer. Besides the impact on speed control and visual workload, we explored how driver's needs were met. Specifically, the following research questions were addressed:

Mental effort, learning effort, task difficulty and safety

- Is the subjective mental effort reduced as it would be due to skill-based behaviour?
- Is the learning effort of the new distance-speedometer acceptable?
- Are the perceived task difficulty and perceived safety as good as with the well-known speedometer?

Usefulness and satisfaction

- Is the system useful and satisfying?

Seriousness and fun

- Is the distance-speedometer's hedonic quality comparable to a normal speedometer?

Speed Control and Visual Workload

- How well do drivers control their speed with the distance-speedometer?
- How fast do drivers perform required speed changes?
- Is the visual workload reduced by the distance-speedometer?

Gaze behaviour

- How fast can the speed information be processed and how much cognitive processing is involved?
- Does the distance-speedometer allow for peripheral guidance?

2. Method

We compared driving with the distance-speedometer to driving with a standard speedometer. As Fig. 2 indicates, to make it comparable to the distance speedometer, a recommended speed was also shown on the standard speedometer by marking limits on its scale.

An evaluation was made of following a road with different speed limits (sign-following), as well as with car-following, which included speed changes of the car ahead at different speeds. In both these separate scenarios the effects of the two HMIs were examined. The research complied with the German Psychological Societies' adaptation of the American Psychological Association's Code of Ethics and was approved by the Institutional Review Board of the TU Braunschweig. Informed consent was obtained from each participant.

2.1. Experimental design

A within-subject design was used. Each driver tested both the distance-speedometer and the standard speedometer for the separate scenarios of sign-following and car-following. A direct comparison of the two scenarios was not sought as the HMIs changed together with the scenarios (the bracket was introduced with the sign-following scenario). Additionally, parameters referencing the car ahead could only be computed for the car-following scenario. The order of both HMIs was counterbalanced, and the order of the two scenarios was also counterbalanced.

2.2. Participants

In order to reduce the risk of underpowered results, a priori sample size calculation was performed using G*Power (Faul, Erdfelder, Lang, & Buchner, 2007). To detect medium effects larger than 0.5, a sample size of $N = 54$ is needed to achieve a power of 0.95 in a t -test for repeated measurements based on $\alpha = 0.05$. In the end, $N = 53$ participants were recruited. Technical problems that caused four broken data files meant that $n = 49$ participants remained in the final sample (see Table 1).

Drivers were randomly selected from an internal database of volunteers and recruited by telephone. They had either normal vision or else they used corrective lenses. The 18 females had a mean age of 30 years ($SD = 15$), while the 31 males had a mean age of 40 years ($SD = 19$). Most of the participants drove three to nine thousand kilometres per year.



Fig. 2. Standard speedometer with a bracket indicating the recommended speed while sign-following. While car-following the bracket was excluded. The speedometer was located at the head-down display.

Table 1
Demographics.

Sample	n	Classification of the yearly mileage (categories represent km in thousands)						Age		Years licensed	
		<3	3–9	9–12	12–20	20–30	30–50	M	SD	M	SD
Female	18	3	12	1	1	1	0	30	15	13	15
Male	31	4	8	6	7	4	2	40	19	22	18

Note. M = mean, SD = standard deviation.

How do you estimate the learning effort to use the system?

very low			low			moderate			high			very high		
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

How difficult was it to follow the speed requirements / the car ahead?

very difficult			difficult			moderate			easy			very easy		
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

How safe did you feel when using system?

very unsafe			unsafe			moderate			safe			very safe		
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

Fig. 3. 15-point scales for measuring learning effort, task difficulty and subjective feeling of safety (translated from German; ratings on task difficulty were inverted).

2.3. Measures

2.3.1. Mental effort, learning effort, task difficulty and safety

Given that mental effort was intended to be reduced in a skill-based manner, we asked for it by using the 220 mm long visual-analogue SEA scale (“Skala zur Erfassung subjektiv erlebter Anstrengung”; Eilers, Nachreiner, & Hänecke, 1986), a German translation of the SMEQ (“Subjective Mental Effort Questionnaire”; Zijlstra & Van Doorn, 1985).

The learning effort, task difficulty and subjective feeling of safety were evaluated using 15-point scales based on a two-step rating system from Heller (1982). As Fig. 3 indicates, the scale contained five verbal categories, with three subcategories to refine the rating.

The HMI’s novelty meant that learning effort and subjective safety could become a great barrier to adoption. Given that the distance-speedometer aims at skill-based control, we sought to assess the subjective rating of difficulty in controlling the speed while using it.

2.3.2. Usefulness and satisfaction

In order to be accepted and frequently used, a system’s functionality must be understood and valued by its users (Nilsson, Várhelyi, & Adell, 2017). We therefore used the Van der Laan Scale (Van der Laan, Heino, & de Waard, 1997) to measure usefulness and satisfaction.

2.3.3. Seriousness and fun

As an extension of the Van der Laan Scale, the hedonic component was measured using two single item 5-point scales that asked for fun and seriousness with anchor points of “serious-playful”, and “fun-boring”. The hedonic component was of interest as the distance-speedometer is not intended to be too playful, but should be fun to use.

2.3.4. Speed control and visual workload

The simulator recorded using a sampling rate of 100 Hz. Driver behaviour (gas pedal) and the resulting behaviour of the vehicle (e.g., acceleration) were recorded and aggregated to mean values for every meter of the measuring section. Speed deviations were computed as the difference between the ego-speed to either the current speed limit or the leading car’s speed. The standard deviations of these speed deviations were calculated as the parameter of interest, which reflected the overall driving quality in terms of speed control. Smaller standard deviations indicate a more homogeneous following behaviour. In addition, gas pedal usage time (percentagewise) was used as an indicator of the effort used to control speed.

For car-following, the mean THW was computed to examine whether it was effectively influenced by the distance-speedometer. In addition, the minimal time to collision (TTC) was computed as a more risk-based measure for the assessment of near rear-end collisions.

In order to discover how fast the drivers reacted to speed signs, we examined at which distance relative to the speed sign the drivers began to accelerate or decelerate (depending on the direction of change). Similarly, beginning with the speed change of the lead car, it was measured how many meters it took for the drivers to begin to accelerate or decelerate.

A secondary task was used to assess the visual workload. The surrogate reference task is a simple visual search task (Mattes & Hallén, 2009), that presented stimuli on a touch screen located to the right of the steering wheel. The subjects reacted by touching the screen. They were instructed to handle as many trials as possible while still driving safely at their discretion. The number of trials completed was used as an inverse indicator of visual workload. The fewer trials were completed, the larger the visual workload in the main driving task was assumed to be. Over 99% of the completed tasks were

completed correctly in every scenario. The task's difficulty level was set to easy as described by the International Organization for Standardization (ISO/DIS 26022, 2007). Since the surrogate reference task requires the same resources as driving (visual, manual), according to Wickens (2002) model of multiple resources, interferences should be likely, causing a performance degradation. When dual tasks are performed, the working memory model from Baddeley (2003) suggests, that if those two tasks share the same working memory resource, performance in one or both tasks deteriorate (Jamson & Merat, 2005). Since in theory, skill-based behaviour does require little working memory compared to rule-based behaviour (Salvendy, 2012), a condition where the distance speedometer allows for skill-based behaviour should interfere less with the reference task than a condition where the normal speedometer demands rule-based behaviour.

2.3.5. Gaze behaviour

To substantiate possible findings, gaze behaviour was recorded as well. Mean glance duration was used as a measure of cognitive processing: the longer a glance, the more cognitive processing is involved due to the task's difficulty (Victor, Harbluk, & Engström, 2005). This was done for the time needed to process the secondary task, the driving task, as well as the time needed to process the speed information which was embedded in the driving task.

For the secondary and the driving task, the mean glance durations could be used directly. The driving task included the head-down-display (where optionally the speedometer was presented) and the street (where optionally the distance speedometer was presented). The secondary task covered the monitor of the surrogate reference task. Fig. 4 shows the simulator setting, where the areas of interest were aligned with.

The time needed to process the speed information could be directly derived in the condition where the speed was shown in the speedometer. There, just the glance time to the speedometer was used.

For the head-up condition, glances to the distance-speedometer could not be separated directly from glances on to the road. Therefore, first the time needed to process the scenery was estimated from the speedometer condition (there was no additional head up information), and then subtracted from the time needed to process the scenery and the overlaying distance speedometer in the distance-speedometer condition. This difference then was used as an approximation for the time needed to process the speed information given by the distance-speedometer.

With respect to the driving task, the duration of glances to the secondary task might also reflect the use of peripheral cues instead of focal cues. If peripheral cues could be used for speed control, one could also extend glance durations to the secondary task. Therefore, longer glances to the secondary task might indicate a greater usage of peripheral cues for speed control.

2.4. Apparatus

The fixed-base half-vehicle mockup simulator of the Department of Engineering and Traffic Psychology at the TU Braunschweig consists of a mock-up cockpit and an array of three screens (2×2 m), in addition to corresponding LCD projectors (1920×1080 pixels each) that cover a 180° field of vision from the driver's seat, which is approximately 2 m away from the front screen's centre. One 7" colour LCD screen (1280×768 pixels) was mounted at the speedometer's typical position. Two identical 7" screens were used as side mirrors, but no rear-view mirror was included. The HUD was simulated on the front screen. The steering wheel, gas pedal and brake pedal were equipped with force feedback. An automatic gearbox was simulated, and driving sounds were played by a surround-sound system. SILAB 4.0 (WIVW GmbH, <https://wivw.de/en/silab>) was used as simulator software.

In order to compute the distances shown in the distance-speedometer (speed in km/h * THW in s = distance in m), it was necessary to define a study specific THW. Therefore, 10 drivers participated in calibration runs (mean age in years = 26 [SD = 10], mean years licensed = 10 [SD = 10], yearly mileage mostly below 3000 km, 6 female). They drove in the car-following scenario with the same instruction and setting as used in the study. An average of 1.7 s THW resulted. Comparable values have been reported in literature (Colbourn, Brown, & Copeman, 1978; Siebert, Oehl, Bersch, & Pfister, 2017).

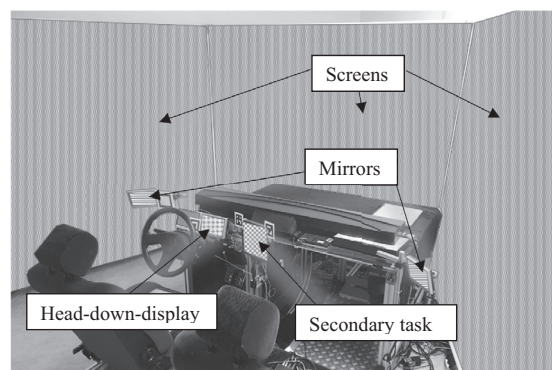


Fig. 4. Configuration of the fixed-base half-vehicle mockup simulator.

Table 2

Sequence of prescribed speed limits (km/h) within the measuring section (indicated by signs or the car ahead).

no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
km/h	50	30	40	50	40	30	40	30	50	40	50	30	30	50	40	30	40	50

2.5. Experimental trial

The car-following and sign-following scenarios used a completely straight track (a rural road, see Fig. 1). In both scenarios there were 18 speed change points within the measuring section, which were located within a distance of 250 m. The speeds were set at 30, 40 and 50 km/h. All possible speed changes were implemented and the same order was used for all the drivers (the effects of certain speed limits were not analysed). The speed change points were either explicitly indicated by signs (sign-following) or implicitly indicated by the programmed speed changes of the car ahead, without using braking lights (car-following). Table 2 indicates the sequence of prescribed speeds (which were indicated either by signs or by the car ahead). Before the measuring section, a training section (30, 40, 50, 40 and 30 km/h, followed by a stop sign) was introduced. Thus, the drivers had time to get comfortable with the combination of the HMI and the scenario. After this, the drivers had to accelerate to 30 km/h before the measuring section started. The final stop sign was set 250 m behind the end of the measuring section.

2.6. Procedure

The participants were first informed about the aim of the study, namely, that it was intended to examine a new kind of interface. They were instructed to drive safely but at a good pace, and to keep up with the speed limits. Nevertheless, it was emphasized that they should drive in the same manner as they normally drove. In addition, they were encouraged to do as many of the surrogate reference task trials as safely possible. Moreover, their overall engagement in both tasks was reinforced by a monetary reward, which was earned by driving fast safely, in addition to completing more secondary tasks. However, they were also told that the reward would be reduced in the event of an accident, or if they were caught by a simulated speed control.

The drivers were then introduced to the driving simulator by means of a short training session. This was followed by the experiment. Each scenario (either car-following or sign-following using either of the two HMIs) started with a description of the HMI used and took about 15 to 20 min, which included a subsequent questionnaire. The descriptions contained a picture of the scenario and the HUD, plus a few sentences explaining the visualization of the set value, the control variable and the optimal match between both. In the case of car-following without the distance-speedometer, the drivers were instructed to just follow as it seems appropriate to them, but at a good pace. For ethical reasons, each driver received the same amount of money and a debriefing. The whole experiment took about 1.5 h.

2.7. Data analysis

Everything was computed using R Statistics 3.4.3 (<https://www.r-project.org/>), Processing 3.3.6 (<https://processing.org/>) and SPSS Statistics 23 (<https://ibm.com/spss>). Average imputation was used for eight of the 3626 questionnaire ratings, which meant that missing ratings were filled in using the average of the corresponding measure of all other participants. For each of the independent scenarios, the two HMIs were tested against each other using a *t*-test for repeated measurements. All statistical tests were performed on a 5% significance level and *p*-values below 0.1 were seen as tendencies. Given the explorative research aim of this study, tests were performed and reported two sided and without Bonferroni correction. That means, we used the alpha level of 5% for all tests in order to explore where differences between the two HMIs are found, which are very unlikely due to chance. However, we also performed a Bonferroni-Holm Correction for multiple comparisons to validate our findings. Comparisons for the sign-following and car-following scenario were corrected separately. While all significant results remained significant after the correction, the reported tendencies lost their significance.

3. Results

3.1. Mental effort, learning effort, task difficulty and safety

Ratings on mental effort lie around the verbal anchor of “some effort” and showed no significant differences (see Table 3). However, there was a tendency in the sign-following scenario that the distance-speedometer was rated as being less demanding.

For the parameters learning effort, task difficulty, and safety, no significant difference between the two HMIs could be shown either in car-following or in sign-following (see Table 4). However, as a tendency, the distance-speedometer took a bit more learning effort in the sign-following scenario than the standard speedometer.

Table 3
Mental effort.

Dependent Variable	Scenario & HMI		M (SD)	t(48)	p	d
Mental effort (220 mm visual-analogue scale)	Sign	Speedometer	89.45 (41.60)	1.75	0.09	0.30
		Distance-speedometer	77.43 (37.60)			
	Car	Speedometer	87.20 (48.03)	−0.39	0.70	0.06
		Distance-speedometer	90.04 (46.17)			

Note. M = mean, SD = standard deviation, d = Cohen's d.

Table 4
Learning effort, task difficulty and safety.

Dependent Variable	Scenario & HMI		M (SD)	t(48)	p	d
Learning effort (15-point scale)	Sign	Speedometer	4.53 (2.67)	−1.78	0.08	0.25
		Distance-speedometer	5.16 (2.37)			
	Car	Speedometer	5.22 (3.61)	−0.41	0.68	0.86
		Distance-speedometer	3.00 (0.43)			
Task difficulty (15-point scale)	Sign	Speedometer	6.76 (2.39)	−0.111	0.27	0.17
		Distance-speedometer	6.35 (2.45)			
	Car	Speedometer	7.39 (2.67)	0.53	0.60	0.10
		Distance-speedometer	7.65 (2.80)			
Safety (15-point scale)	Sign	Speedometer	9.94 (3.02)	−0.73	0.47	0.11
		Distance-speedometer	10.24 (2.40)			
	Car	Speedometer	9.65 (2.74)	−0.04	0.97	1.12
		Distance-speedometer	9.67 (2.59)			

Note. * = $p < 0.05$, M = mean, SD = standard deviation, d = Cohen's d.

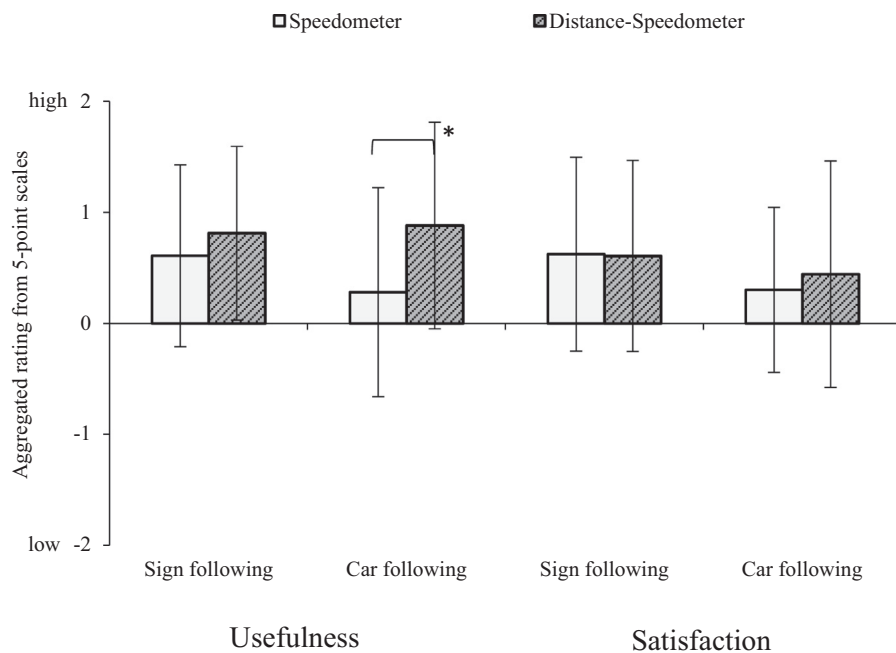


Fig. 5. Usefulness and satisfaction for the two HMIs in the two scenarios (mean and standard deviation). Both dimensions were rated on several 5-point scales [−2 up to +2] (mean and SD). (* = $p < 0.05$).

3.2. Usefulness and satisfaction

As Fig. 5 shows, the perceived usefulness was significantly higher for the distance-speedometer in car-following but not in sign-following. The mean satisfaction did not differ between the two HMIs in either scenario (see Table 5). Overall, all values were in the medium positive range of the two scales.

Table 5
Usefulness and satisfaction.

Dependent Variable	Scenario & HMI		M (SD)	t(48)	p	d
Usefulness (5-point scales [−2 up to +2])	Sign	Speedometer	0.61 (0.82)	1.69	0.10	0.25
		Distance-speedometer	0.81 (0.78)			
	Car	Speedometer	0.28 (0.49)	3.12	<0.01*	0.77
		Distance-speedometer	0.88 (0.94)			
Satisfaction (5-point scales [−2 up to +2])	Sign	Speedometer	0.62 (0.87)	−0.13	0.90	0.01
		Distance-speedometer	0.61 (0.86)			
	Car	Speedometer	0.30 (0.74)	−0.78	0.44	0.18
		Distance-speedometer	0.44 (1.02)			

Note. * = $p < 0.05$, M = mean, SD = standard deviation, d = Cohen's d.

3.3. Seriousness and fun

There was a significantly higher rating for the distance-speedometer with regard to fun (vs. boring) in both scenarios (see Table 6). In car-following, the speedometer was rated as being more serious (vs. playful) than the distance-speedometer (see Fig. 6).

3.4. Speed control and visual workload

While car-following, the standard deviation of speed deviations was significantly greater with the speedometer than with the distance-speedometer (see Table 7 and Fig. 7). Thus, drivers with the distance-speedometer followed the speed of the car

Table 6
Seriousness and fun.

Dependent Variable	Scenario & HMI		M (SD)	t(48)	p	d
Seriousness (5-point scale)	Sign	Speedometer	3.69 (0.92)	−3.30	<0.01*	0.58
		Distance-speedometer	3.10 (1.12)			
	Car	Speedometer	3.45 (1.02)	−0.53	0.60	0.12
		Distance-speedometer	3.33 (1.07)			
Fun (5-point scale)	Sign	Speedometer	2.90 (0.68)	4.12	<0.01*	0.74
		Distance-speedometer	3.45 (0.82)			
	Car	Speedometer	2.65 (0.83)	3.83	<0.01*	0.78
		Distance-speedometer	3.33 (0.92)			

Note. * = $p < 0.05$, M = mean, SD = standard deviation, d = Cohen's d.

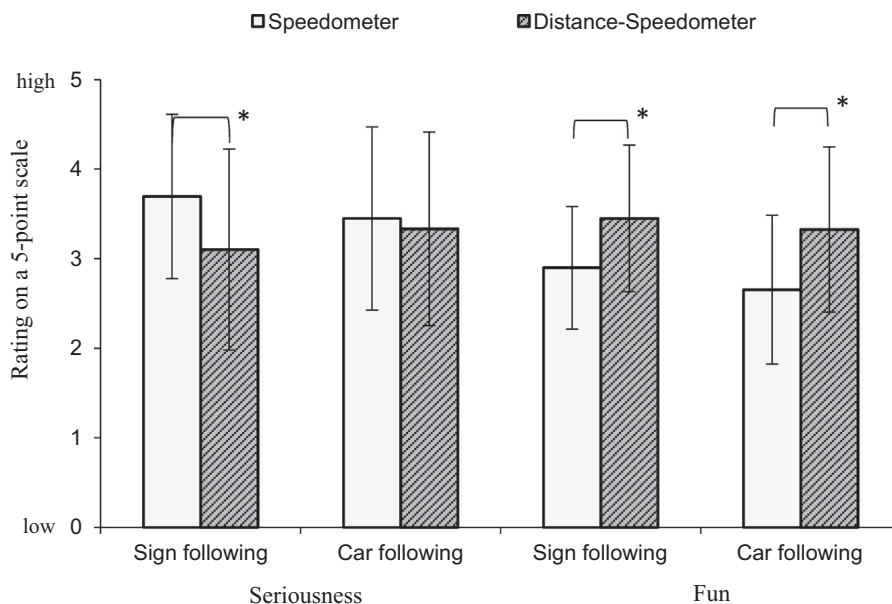


Fig. 6. Seriousness and fun, rated on 5-point scales (mean and SD). (* = $p < 0.05$).

Table 7

Driving performance and visual workload.

Dependent Variable	Scenario & HMI		<i>M</i> (<i>SD</i>)	<i>t</i> (48)	<i>p</i>	<i>d</i>
Standard deviation of speed deviations (in km/h)	Sign	Speedometer	3.89 (0.82)	1.52	0.13	0.45
		Distance-speedometer	3.73 (0.75)			
	Car	Speedometer	5.29 (0.86)	6.78	<0.01*	1.25
		Distance-speedometer	4.52 (0.66)			
Handled tasks per meter (n)	Sign	speedometer	0.03 (0.01)	−4.86	<0.01*	0.53
		Distance-speedometer	0.04 (0.01)			
	Car	speedometer	0.04 (0.01)	1.13	0.26	0.64
		Distance-speedometer	0.04 (0.01)			
Gas usage time (percentage)	Sign	Speedometer	49.18 (6.17)	0.31	0.76	0.26
		Distance-speedometer	49.04 (5.64)			
	Car	Speedometer	42.54 (6.17)	−2.95	<0.01*	0.66
		Distance-speedometer	44.32 (5.98)			

Note. * = $p < 0.05$, *M* = mean, *SD* = standard deviation, *d* = Cohen's *d*.

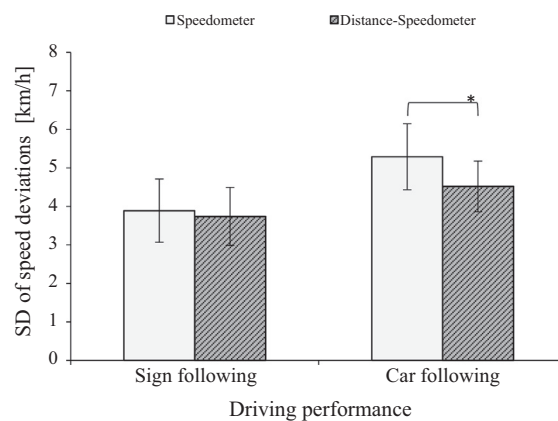


Fig. 7. Driving performance with both HMIs is shown as the standard deviation of speed deviations (mean and *SD*) for sign-following and car-following. (* = $p < 0.05$).

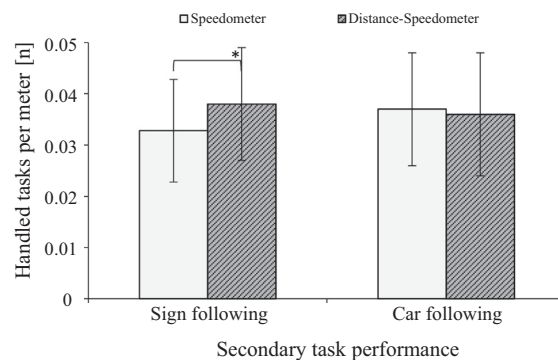


Fig. 8. Surrogate reference task performance (number of tasks per meter) while car-following and sign-following with the two HMIs (mean and *SD*). (* = $p < 0.05$).

ahead more closely. Accordingly, the gas pedal was used significantly more often with the distance-speedometer (see Table 7). However, in this scenario, the engagement in the secondary task did not differ significantly between the two HMIs (see Fig. 8).

For sign-following, a different pattern was found. The standard deviation of speed deviations did not differ significantly between the two HMIs. However, drivers handled significantly more secondary tasks with the distance-speedometer (see Table 7), leading to one completed task every 26.3 m in comparison to one completed task every 30.3 m. Again, Figs. 7 and 8 show these effects. The frequency of gas pedal usage did not differ between the two HMIs.

Table 8

Deceleration and acceleration.

Dependent Variable	Scenario & HMI		<i>M</i> (<i>SD</i>)	<i>t</i> (48)	<i>p</i>	<i>d</i>
Acceleration (Relative distance in m)	Sign	Speedometer	−3.05 (16.43)	1.12	0.27	0.40
		Distance-speedometer	−6.23 (16.96)			
	Car	Speedometer	22.58 (21.56)	−0.32	0.75	0.26
		Distance-speedometer	21.50 (17.26)			
Deceleration (Relative distance in m)	Sign	Speedometer	−3.19 (18.59)	1.09	0.06	0.51
		Distance-speedometer	−10.35 (19.52)			
	Car	Speedometer	39.44 (24.08)	3.01	<0.01*	0.67
		Distance-speedometer	28.49 (13.79)			

Note. * = $p < 0.05$, *M* = mean, *SD* = standard deviation, *d* = Cohen's *d*.

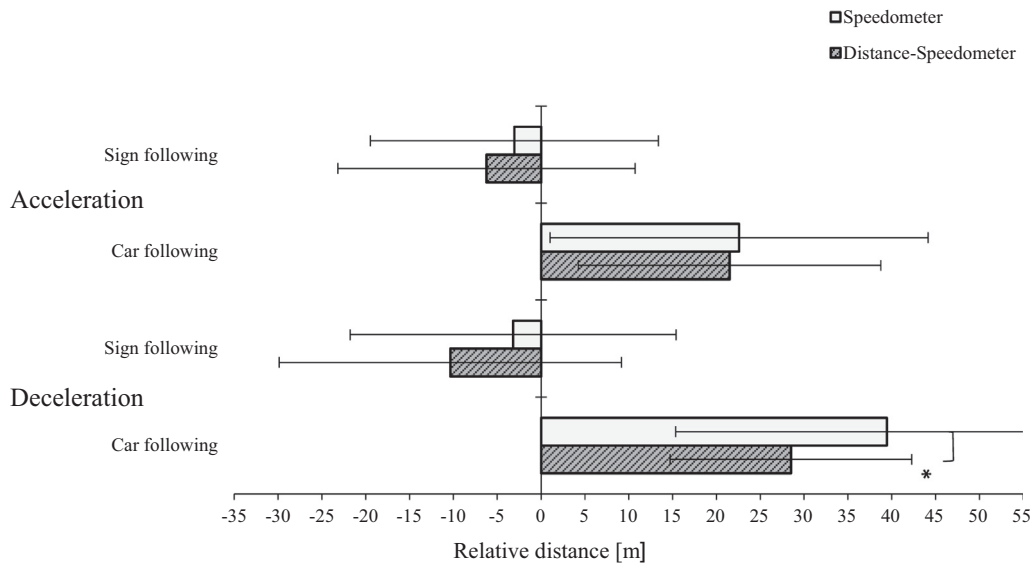


Fig. 9. Relative distance of the deceleration and acceleration of the car, measured from the point where the speed limit sign was or the lead car began to change speed (mean and SD). (* = $p < 0.05$).

In order to examine car-following behaviour with the HMIs more closely, the mean THW and minimum TTC were examined. With the distance-speedometer, the THW was significantly shorter, $t(48) = 2.37$, $p = 0.02$, $d = 0.58$. The mean THW with the speedometer was 2.31 s ($SD = 0.83$) and 2.05 s ($SD = 0.33$) with the distance-speedometer.

Regarding the minimal TTC no significant difference could be shown, $t(48) = 1.57$, $p = 0.12$, $d = 0.46$. On average, the TTC was 3.54 s with the speedometer ($SD = 1.89$ s) and 3.07 s with the distance-speedometer ($SD = 1.54$ s).

Additionally, the point of acceleration and deceleration was examined relative to the signs and the changing behaviour of the car ahead. Table 8 and Fig. 9 depict the findings. There was a significant difference for the deceleration in car-following. With the distance-speedometer drivers decelerated significantly earlier after the preceding car's deceleration. The same tendency could be found for sign-following.

3.5. Gaze behaviour

Due to the configuration of the eye-cam, only 41 Participants could be analysed with regard to their gaze behaviour. For car-following, the mean glance duration on the driving task and the secondary task showed no significant differences. The time needed to process the speed information showed no significant differences as well (see Table 9).

For sign-following, a different pattern was found. While the mean glance duration on the driving task and the secondary task did not differ, the mean time needed for getting the speed information was significantly shorter with the distance-speedometer (see Table 9).

Table 9
Glance patterns.

Dependent Variable	Scenario & HMI		<i>M</i> (<i>SD</i>)	<i>t</i> (40)	<i>p</i>	<i>d</i>
Mean glance duration on the driving task	Sign	Speedometer	1.00 (0.55)	−0.62	0.54	0.09
		Distance-speedometer	0.94 (0.49)			
	Car	Speedometer	0.92 (0.47)	1.61	0.12	0.25
		Distance-speedometer	1.12 (0.61)			
Mean glance duration on the secondary task	Sign	Speedometer	0.48 (0.19)	−0.10	0.92	0.15
		Distance-speedometer	0.45 (0.17)			
	Car	Speedometer	0.48 (0.16)	−0.98	0.33	0.01
		Distance-speedometer	0.48 (0.14)			
Mean time for processing the speed information	Sign	Speedometer	0.59 (0.23)	−4.70	<0.01*	0.75
		Distance-speedometer	0.33 (0.28)			
	Car	Speedometer	0.36 (0.28)	−0.45	0.66	0.07
		Distance-speedometer	0.31 (0.66)			

Note. * = $p < 0.05$, *M* = mean, *SD* = standard deviation, *d* = Cohen's *d*.

4. Discussion and conclusion

This paper contributes to the research on human factors by designing and evaluating an ecological HMI for speed and distance control. The distance-speedometer was developed in order to enable drivers to improve the perception and control of their own speed, in comparison to both the speed limit and a safe THW. Literature on the design of ecological interfaces is extended by applying Rasmussen's model of human behaviour and the principles of ecological interface design to a new measure for traffic management. In order to be less demanding, the distance-speedometer was intended to trigger skill-based behaviour. To ensure efficient traffic management, drivers should stick to the speed and distance recommendations provided. Whether the ecological interface can substitute a standard speedometer was explored in a driving simulator study.

Although substantial additional new visual information was provided by the distance-speedometer, the subjective evaluation of task difficulty, and perceived safety showed no disadvantages under both driving scenarios.

The tendency of less learning effort involved in the standard speedometer under sign-following can be explained by its familiarity. The tendency that the distance-speedometer under sign-following was rated as less demanding, is in keeping with the finding that more secondary tasks are conducted while using it. When using the distance-speedometer while sign-following, the drivers performed substantially more secondary tasks and this indicates a reduction in visual workload caused by the driving task.

With regard to driving behaviour, during sign-following, speed relative to the speed limit was controlled in a comparable manner with both HMIs. Nevertheless, there was a tendency for the drivers to decelerate earlier with the distance speedometer, which indicates an increased compliance with the speed limits. Additionally, for sign following, the mean glance time needed for getting the speed information was significantly shorter with the distance-speedometer. This also provides evidence for the theory that that skill-based processing of the speed information is enabled. However, the time needed to look at the speedometer could be partly prolonged due to having to shift the glance and start the processing at a new spatial location. Therefore, a follow up study might present the normal speedometer in the head-up-display as well. Additionally, a cognitive loading secondary task could be used to validate that the distance-speedometer triggers skill-based behaviour and thus interferes less with such task.

During car-following, the distance-speedometer enabled the drivers to more precisely adapt their speed to the car ahead. This was achieved by more frequent gas pedal usage, as well as by significantly faster reactions to the car's speed changes. However, it did not lead to additional visual workload, as the results of the secondary task indicate. Although the drivers controlled their speed better with the distance-speedometer, they performed a comparable number of secondary tasks. That free focal capacities are used for speed control rather than for the secondary task can be explained by the fact that the allocation of resources is task dependent and the driving task is more important during car-following. Since also the car in front constitutes some kind of ecological interface, more precise driving behaviour is enabled with the same processing time. The distance-speedometer provides drivers with a visual stimulus for the distance, which does not add up on the glance time and therefore might be processed automatically, indicating skill-based behaviour.

For the car-following scenario, the perceived usefulness of the distance-speedometer was rated better than the perceived usefulness of the speedometer. For sign following, no significant differences were found. Furthermore, the distance-speedometer appeared to be more fun for the drivers under both scenarios. It was only with regard to sign-following that the seriousness of the standard speedometer significantly exceeded the level of the distance-speedometer's seriousness.

On the basis of both behavioural effects and also its acceptance and hedonic qualities, the new HMI can be recommended, as the physical terms of speed seem dispensable. Based on the hedonic quality, the distance-speedometer might be perceived by customers as helpful and not as patronizing. This would constitute a big advantage against technology like intelligent speed adaptation systems, which in fact do patronize.

The distance speedometer's usage for traffic management should be examined on a larger scale. Nevertheless, this study showed that the distance-speedometer has the potential to systematically influence the drivers' behaviour. In the car-following scenario, the drivers had a mean THW that was shorter and therefore more in keeping with the predetermined THW visualized by the distance-speedometer, while the minimal TTC showed no difference. This indicates that the drivers generally stuck to the computed distance, while rear-end collisions remained as unlikely as while driving with the normal speedometer. Regarding the THW, a similar effect was found for the sign-following task. Drivers with the distance-speedometer decelerated in keeping with the recommendation provided by the bracket, and achieved comparable results to the standard speedometer.

For the assumption that the distance speedometer can be processed peripheral, no indication was found. The glances to the secondary task were comparable in their mean duration. If they would have been longer while using the distance-speedometer, one could have argued that thereby simultaneously driving could be handled by peripheral cues, but this was not the case.

Finally, it should be noted that the existing legislation requires a numerical speedometer which matches the required speed format of a given country. For example, the German legislation even prohibits the use of speedometers which only shows miles per hour instead of kilometres per hour (*Straßenverkehrs-Zulassungs-Ordnung*, 2012). To avoid legal problems while using the distance-speedometer, it could be complemented by a legal speedometer in the head-down display. In that case, interferences between both types of speedometer should be further examined. Furthermore, innovations, especially those which increase traffic safety, may even initiate a modification of the current legislation.

5. Outlook

As it was out of the scope of this study, further research should verify how much of the effects can be explained by the position of the HUD in comparison to the speedometer behind the steering wheel. Other limitations which should be further examined, are the HMI's impact on the situation awareness, as well as interactions with different driving scenarios. Effects of augmentation failures during cornering should be investigated.

The distance-speedometer could further enhance traffic safety and efficiency when it is used to decompress or compress the traffic (by defining the THW) and to increase or decrease the vehicle's speed. To meet individual needs, e.g. due to reduced responsiveness, individual preferences of speed and THW should be integrated. The expected safety gain should be quantified for real world scenarios.

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Declaration of Competing Interest

None.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.trf.2019.05.012>.

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